

AGRISMART AI

SMART IRRIGATION PREDICTION · SIH 2026 · BONUS MODULE B · ONE-PAGE TECHNICAL MODEL REPORT

01 TASK & SIH ALIGNMENT

Objective: Predict the irrigation state from environmental, crop, soil and growth-stage information, then translate the prediction into a practical irrigation action. This addresses SIH Bonus Module B: irrigation prediction from soil moisture, weather forecast, crop type and growth stage, with model/logic and validation reported.

0.876	95.56%	0.956	0.0078 ms
UNSEEN-COMBINATION MACRO-F1	UNSEEN-COMBINATION ACCURACY	WEIGHTED F1	INFERENCE TIME

02 DATASET & VALIDATION

Dataset: 16,411 raw records → 16,283 clean after 128 duplicates removed. Three classes: 8,934 / 6,227 / 1,122. Data is explicitly synthetic, with repeating temperature cycles and a −0.9761 temperature–humidity correlation. Validation: Stratified Random Holdout + Unseen Combination Evaluation. Grouped unseen crop–soil–stage evaluation is authoritative because random splitting can be overly optimistic.	Selected model: XGBoost using the frozen six-feature production pipeline (OneHot + Scaler + XGBoost). Screening included majority/rule baselines, Logistic Regression, Random Forest, HistGradientBoosting and XGBoost. Inputs: crop, soil type, growth stage, soil moisture, temperature and humidity. Weather forecast context is applied by the recommendation layer.
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03 MODEL / APPROACH

04 GENERALIZATION RESULTS

CLASS	PRECISION	RECALL	F1	BENCHMARK	MACRO-F1
0 — No irrigation	0.979	0.979	0.979	Majority baseline	0.2362
1 — Irrigation required	0.977	0.969	0.973	Rule from MOI + Env	0.4401
2 — Excess water	0.661	0.693	0.677	Logistic Regression	0.6555
				Random holdout — XGBoost	0.9982
				Unseen combinations — XGBoost	0.8760

Interpretation: Random-holdout Macro-F1 is 0.9982, but grouped unseen-combination Macro-F1 is 0.8760. The 12.22-point generalization gap is attributed to deterministic/repeating patterns; the unseen-combination result is therefore the safer headline benchmark.

05 EXTERNAL FIELD TRANSFER

External Mendeley data: 3,584 clean records from Iraq, collected with custom IoT sensor nodes over one week. Frozen six-feature production pipeline: 81.86% accuracy , but precision 7.14% , recall 0.85% , F1 0.015 , ROC-AUC 0.582 . Classified as poor transfer, especially for irrigation-positive detection. All predictions were high-confidence, showing that probability is not a guarantee of field correctness.	<table><tr><th>STATE</th><th>ACTION</th></tr><tr><td>Irrigation required</td><td>≥70% rain probability or ≥5 mm forecast → delay; moderate rain → reduce; otherwise irrigate unless confidence is borderline.</td></tr><tr><td>Excess water</td><td>Avoid irrigation; with significant rain, check drainage / waterlogging risk.</td></tr><tr><td>No irrigation</td><td>No irrigation; monitor when confidence is borderline.</td></tr></table>	STATE	ACTION	Irrigation required	≥70% rain probability or ≥5 mm forecast → delay ; moderate rain → reduce ; otherwise irrigate unless confidence is borderline.	Excess water	Avoid irrigation ; with significant rain, check drainage / waterlogging risk.	No irrigation	No irrigation ; monitor when confidence is borderline.
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06 WEATHER-AWARE FARMER LOGIC

07 LIMITATIONS & RESPONSIBLE USE

• Synthetic data contains deterministic sequences and repeating environmental patterns. • Random holdout can overestimate generalization when similar sequence points occur in train and test. • Class 2 (excess water) is substantially harder than Classes 0 and 1. • External transfer is poor, particularly for irrigation-positive detection. • The system does not directly model irrigation volume or a complete physical soil-water balance. • Weather forecasts add uncertainty. **Output is decision support, not an autonomous irrigation controller.**
SIH 2026: Task · Dataset & validation · Model/approach · Metric & result · Baselines · Limitations · Bonus Module B logic/model validation. Primary headline metric is the grouped unseen-combination benchmark.